

COLUMBIA METRO, SC, USA

WMO: 723100

Lat: 33.942N

Lon: 81.118W

Elev: 69

StdP: 100.50

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13883

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.7	-2.8	-14.9	1.0	0.9	-12.1	1.3	3.2	10.8	11.3	9.4	12.2	1.3	260	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	36.2	24.1	34.9	23.8	33.8	23.7	25.8	32.0	25.4	31.5	25.0	30.8	3.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.4	19.5	27.9	23.9	18.9	27.5	23.4	18.4	27.2	80.1	32.3	77.9	31.6	76.0	30.9	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.5	6.5	-8.5	38.2	2.0	1.7	-9.9	39.4	-11.0	40.5	-12.1	41.5	-13.5	42.7		
			-9.4	26.9	1.9	0.7	-10.7	27.4	-11.9	27.8	-13.0	28.2	-14.3	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	7.8	9.8	13.6	18.2	22.6	26.4	27.9	27.4	24.3	18.4	12.4	9.1
	DBStd	8.27	5.49	5.06	4.94	4.14	3.48	2.50	1.96	2.14	3.05	4.30	4.59	5.09
	HDD10.0	302	111	62	22	1	0	0	0	0	0	1	26	79
	HDD18.3	1319	329	242	161	53	7	0	0	0	2	53	184	287
	CDD10.0	3301	42	56	133	248	392	492	556	539	431	262	98	53
	CDD18.3	1276	2	3	13	49	141	243	298	281	182	56	7	2
	CDH23.3	12847	7	24	142	528	1383	2486	3228	2916	1609	462	51	12
	CDH26.7	5629	0	2	23	152	549	1182	1593	1362	649	114	3	0
Wind	WSAvg	2.6	2.8	2.9	3.1	3.2	2.9	2.6	2.5	2.2	2.4	2.2	2.2	2.4
Precipitation	PrecAvg	1266	116	103	125	81	91	122	143	155	93	76	71	88
	PrecMax	1791	235	220	277	174	225	376	444	380	200	296	159	217
	PrecMin	870	21	22	14	7	18	19	14	26	2	0	10	16
	PrecStd	231	53	50	65	46	51	78	95	91	52	75	43	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.2	26.5	29.0	31.8	35.2	37.2	37.9	37.7	35.1	31.5	26.9	24.9
		MCWB	17.9	18.4	18.0	19.8	21.8	23.6	24.3	24.2	22.9	22.8	19.3	18.6
	2%	DB	21.5	23.4	26.9	30.1	33.1	35.5	36.2	35.6	33.6	29.5	24.8	22.5
		MCWB	16.0	16.0	17.3	19.1	21.5	23.6	24.5	24.3	22.8	21.1	17.8	17.6
	5%	DB	19.3	21.2	24.8	28.6	31.6	34.0	34.7	34.2	32.1	27.8	22.9	20.2
		MCWB	15.0	14.7	15.9	18.5	21.2	23.4	24.3	24.3	22.5	20.1	16.8	16.3
	10%	DB	17.0	18.9	22.6	26.6	29.9	32.5	33.5	32.8	30.6	25.9	20.9	18.1
		MCWB	13.5	13.5	14.9	17.4	20.6	23.0	24.2	24.0	22.1	19.4	16.4	14.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	19.8	20.1	22.0	24.4	25.8	26.8	26.5	25.4	24.8	21.7	20.6
		MCDB	22.0	23.3	25.3	27.8	29.3	32.4	33.6	32.8	30.4	28.0	23.9	23.0
	2%	WB	17.7	18.1	18.8	21.1	23.4	25.2	26.0	25.8	24.6	23.5	20.0	18.8
		MCDB	20.4	22.0	23.9	26.6	29.0	31.9	32.3	31.9	29.4	26.9	22.6	21.1
	5%	WB	16.0	16.4	17.8	20.2	22.6	24.6	25.5	25.4	24.1	22.2	18.8	17.2
		MCDB	18.5	19.7	22.7	25.5	28.4	30.9	31.8	31.3	28.6	25.7	21.5	19.6
	10%	WB	14.1	14.7	16.6	19.1	21.9	24.1	25.0	24.9	23.5	20.9	17.3	15.3
		MCDB	16.4	17.8	20.7	24.4	27.5	30.0	31.1	30.3	27.8	24.5	20.3	17.9
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.2	12.8	13.0	12.1	10.9	10.6	10.2	10.5	12.2	12.9	11.6
		MCDBR	13.8	14.7	15.1	14.8	13.7	12.7	12.2	11.9	12.7	13.4	14.7	13.4
		MCWBR	9.2	8.7	7.2	6.0	4.8	3.8	3.5	3.2	4.0	6.1	8.6	8.9
	5% WB	MCDBR	11.4	12.4	12.1	11.6	11.1	10.9	10.7	10.3	9.7	9.5	10.5	11.1
	MCWBR	8.8	8.6	6.9	5.8	4.3	3.8	3.6	3.2	3.5	5.2	7.5	8.5	
Clear-Sky Solar Irradiance	taub	0.324	0.333	0.371	0.412	0.477	0.510	0.556	0.528	0.445	0.402	0.352	0.322	
	taud	2.488	2.461	2.382	2.261	2.127	2.078	1.972	2.037	2.256	2.341	2.441	2.531	
	Ebn at Noon	887	917	903	877	820	790	751	765	818	826	844	870	
	Edh at Noon	86	98	115	135	157	164	182	168	129	109	90	78	
All-Sky Solar Radiation	RadAvg	2.77	3.40	4.48	5.64	6.14	6.39	6.08	5.53	4.77	3.99	3.05	2.42	
	RadStd	0.14	0.36	0.33	0.41	0.48	0.45	0.40	0.27	0.41	0.48	0.24	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		+0.67	+0.67	N/A	N/A	N/A	N/A	-96	+232	+159
(47)	Regional (60 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	+189	+133

Nomenclature: See separate page